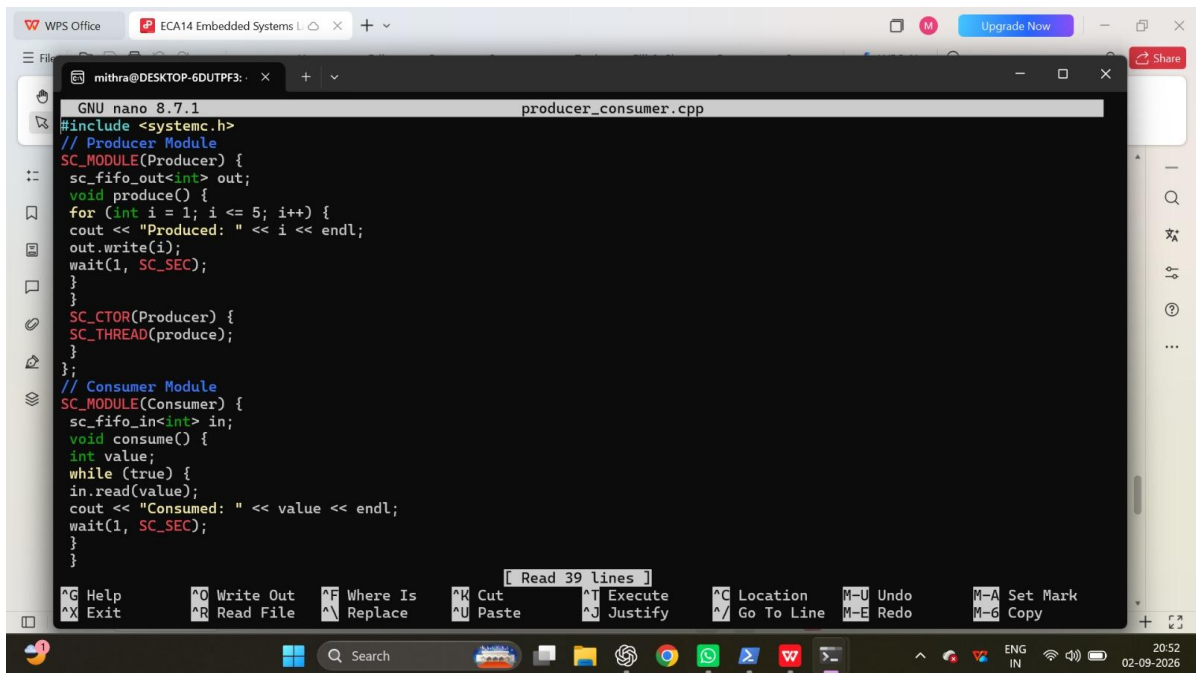


Exp No: 25

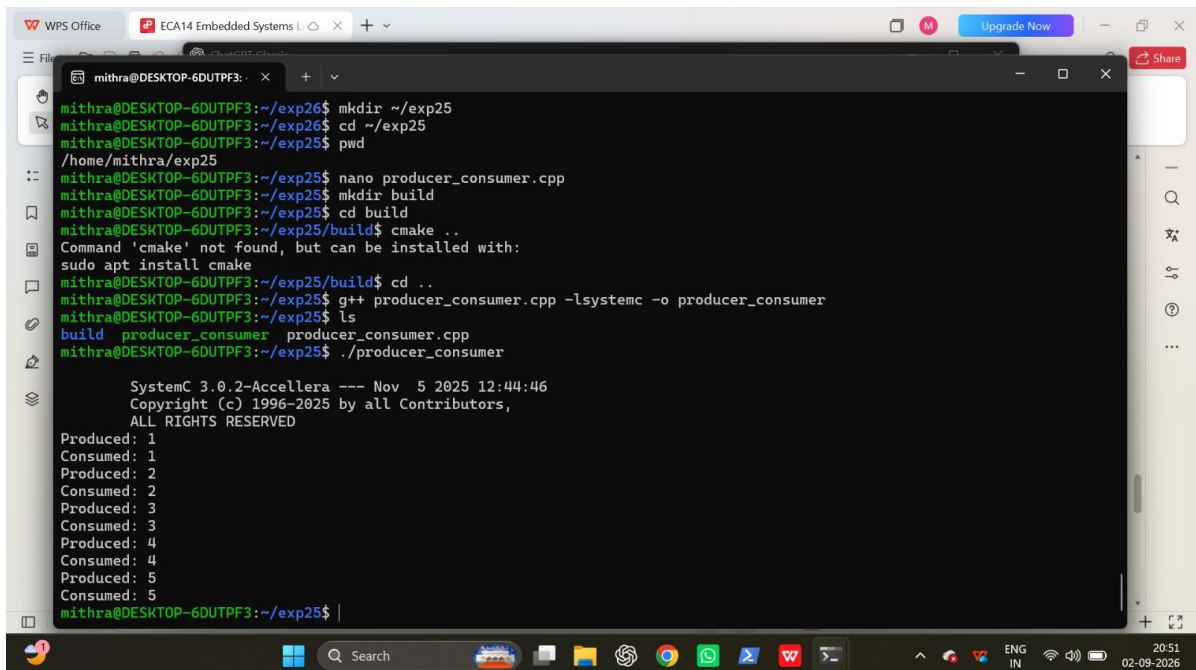
PRODUCER-CONSUMER PROBLEM USING SYSTEMC



The screenshot shows a nano editor window titled 'producer_consumer.cpp' with the following code:

```
GNU nano 8.7.1 producer_consumer.cpp
#include <systemc.h>
// Producer Module
SC_MODULE(Producer) {
    sc_fifo_out<int> out;
    void produce() {
        for (int i = 1; i <= 5; i++) {
            cout << "Produced: " << i << endl;
            out.write(i);
            wait(1, SC_SEC);
        }
    }
    SC_CTOR(Producer) {
        SC_THREAD(produce);
    }
};
// Consumer Module
SC_MODULE(Consumer) {
    sc_fifo_in<int> in;
    void consume() {
        int value;
        while (true) {
            in.read(value);
            cout << "Consumed: " << value << endl;
            wait(1, SC_SEC);
        }
    }
};
```

The bottom status bar of the nano editor shows: [Read 39 lines]



The screenshot shows a terminal window with the following commands and output:

```
mithra@DESKTOP-6DUTPF3:~/exp25$ mkdir ~/exp25
mithra@DESKTOP-6DUTPF3:~/exp25$ cd ~/exp25
mithra@DESKTOP-6DUTPF3:~/exp25$ pwd
/home/mithra/exp25
mithra@DESKTOP-6DUTPF3:~/exp25$ nano producer_consumer.cpp
mithra@DESKTOP-6DUTPF3:~/exp25$ mkdir build
mithra@DESKTOP-6DUTPF3:~/exp25$ cd build
mithra@DESKTOP-6DUTPF3:~/exp25/build$ cmake ..
Command 'cmake' not found, but can be installed with:
sudo apt install cmake
mithra@DESKTOP-6DUTPF3:~/exp25/build$ cd ..
mithra@DESKTOP-6DUTPF3:~/exp25$ g++ producer_consumer.cpp -lsystemc -o producer_consumer
mithra@DESKTOP-6DUTPF3:~/exp25$ ls
build producer_consumer producer_consumer.cpp
mithra@DESKTOP-6DUTPF3:~/exp25$ ./producer_consumer

SystemC 3.0.2-Accellera --- Nov  5 2025 12:44:46
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Produced: 1
Consumed: 1
Produced: 2
Consumed: 2
Produced: 3
Consumed: 3
Produced: 4
Consumed: 4
Produced: 5
Consumed: 5
mithra@DESKTOP-6DUTPF3:~/exp25$
```